

Apache Kafka - Complete Beginner to Advanced Guide

What is Apache Kafka?

Apache Kafka is a distributed event streaming platform used for building real-time data pipelines and streaming applications. It allows producers to publish messages and consumers to process them reliably at scale.

Core Components

Producer, Consumer, Topic, Partition, Offset, Broker, Consumer Group, Controller, and KRaft are the key building blocks.

Topics and Partitions

Topics organize messages. Partitions enable scalability and parallel processing. Ordering is guaranteed within a partition.

Offsets

Each message in a partition has a unique offset. Consumers track offsets to know which records have been processed.

Consumer Groups

Consumers in the same group share partitions, enabling horizontal scaling and fault tolerance.

Replication

Kafka replicates partitions across brokers for high availability. One replica acts as leader while others are followers.

KRaft Architecture

Modern Kafka uses KRaft mode, eliminating ZooKeeper and simplifying cluster management.

`auto.offset.reset`

Determines what a consumer does when no committed offset exists. Values include earliest, latest, and none.

Kafka Connect

Framework for integrating Kafka with databases, cloud storage, and enterprise systems.

Kafka Streams

Client library for stream processing, aggregations, joins, and real-time analytics.

Schema Registry

Manages schemas for Avro, Protobuf, and JSON to ensure producer-consumer compatibility.

Exactly Once Semantics

Kafka supports idempotent producers and transactions to avoid duplicate processing.

Performance Tuning

Tune batch size, linger.ms, compression, partitions, retention policies, and consumer settings.

Monitoring

Monitor consumer lag, broker health, throughput, latency, ISR status, and storage utilization.

Use Cases

Event-driven architecture, microservices communication, log aggregation, analytics pipelines, IoT, and financial systems.

Kafka End-to-End Flow

Producer -> Topic -> Partition -> Broker -> Consumer Group -> Consumer. Messages are written sequentially and consumed using tracked offsets.